LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Unit Test: Structure of the Atom

Chapter / Topic: Structure of the Atom (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.

SECTION A

6 marks

Choose the correct option.

Q1. The electron was discovered by: **[1]**

(a) Rutherford (b) Chadwick
(c) J. J. Thomson (d) Bohr

Q2. The neutron was discovered by: **[1]**

(a) Rutherford (b) Thomson
(c) Chadwick (d) Goldstein

Q3. The maximum number of electrons that can be held in the L shell is: **[1]**

(a) 2 (b) 8
(c) 18 (d) 32

Q4. An element has the electronic configuration 2, 8, 3. Its atomic number is: **[1]**

(a) 3 (b) 8
(c) 11 (d) 13

Q5. Isotopes of an element have the same: **[1]**

(a) number of protons (b) number of neutrons
(c) mass number (d) physical properties

Q6. **Assertion (A):** Isotopes of an element have similar chemical properties.
Reason (R): Chemical properties depend on the valence electrons, which are the same for all isotopes of an element. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Write the number of protons and neutrons in the nuclei of carbon-12 and carbon-14 (Z = 6). **[2]**

OR

What are valence electrons? Find the valency of magnesium (Z = 12).

Q8. What was the main drawback of Rutherford's model of the atom? **[2]**

SECTION C

6 marks

Q9. An atom has 17 protons and 18 neutrons. **[3]**

(i) What is its atomic number? **[1]**

(ii) What is its mass number? **[1]**

(iii) Write its electronic configuration and valency. **[1]**

Q10. Describe Thomson's model of the atom and give one limitation. **[3]**

OR

Define isotopes and isobars with one example each, and give two uses of isotopes.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Describe Bohr's model of the atom. **[3]**

(ii) Write the electronic configurations of the ions Na^+ (Z = 11) and Cl^- (Z = 17). **[2]**

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Natural copper (atomic number 29) is a mixture of two isotopes: copper-63, making up 69% of natural copper, and copper-65, making up 31%. Chemists use the average atomic mass of copper in calculations. **[4]**

(i) What are isotopes? **[1]**

(ii) How many protons does each copper isotope have? **[1]**

(iii) Calculate the average atomic mass of copper. **[2]**

~ All the best! ~